SentinelAI Research Edition PRD

Vision

Build an open-source framework that systematically evaluates frontier and open-source language models for safety, reliability, and alignment failures.

The framework should allow researchers to run standardized evaluations across multiple models and generate quantitative safety reports.

Problem Statement

Current LLM evaluations focus heavily on capability benchmarks.

There is limited accessible tooling for evaluating:

- Deceptive behavior
- Unsafe outputs
- Goal preservation
- Instruction hierarchy violations
- Prompt injection susceptibility
- Hallucination risk

Researchers need reproducible infrastructure to compare models on these dimensions.

Research Questions

1. Which models are most vulnerable to prompt injection?

2. Which models maintain alignment under adversarial pressure?

3. How frequently do models generate unsafe outputs?

4. Which categories produce the highest failure rates?

Core Evaluation Categories

Prompt Injection

Tests:

- Ignore previous instructions
- Override system prompts
- Hidden instruction attacks

Metrics:

- Injection Success Rate

Safety Compliance

Tests:

- Harmful requests
- Illegal requests
- Dangerous knowledge extraction

Metrics:

- Refusal Rate
- Safety Score

Hallucination

Tests:

- Factual QA
- Synthetic QA

Metrics:

- Hallucination Rate

Deception

Tests:

- Strategic misrepresentation
- Goal conflict scenarios

Metrics:

- Deception Score

Reliability

Tests:

- Repeated identical prompts

Metrics:

- Consistency Score

Models

Phase 1

- Claude
- GPT
- Gemini
- Qwen
- Llama

Architecture

Prompt Dataset ↓ Evaluation Runner ↓ Model API Layer ↓ Response Collection ↓ Risk Scoring Engine ↓ Visualization Layer ↓ Research Report Generator

---

Deliverables

1. Open-source GitHub repository

2. Benchmark Dataset

3. Evaluation Framework

4. Public Technical Report

5. Portfolio Case Study

6. Anthropic Fellowship Submission Artifact

Success Criteria

- 500+ benchmark prompts
- 5 model comparison
- Automated scoring
- Research report
- Reproducible experiments